



K. K. Wagh Institute of Engineering Education & Research
Department of Computer Engineering

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Project Based Learning

Project Group No. : 3

Software Requirements Specification (SRS)

CIVICORBIT: AN AI-POWERED GAMIFIED PLATFORM FOR CIVIC ENGAGEMENT TRACKING

(Implemented as “CivicMitra”)

Team Members

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- **Functional Requirements**

ID	Requirement	Description
FR-1	Evidence Submission	User shall be able to submit proof of civic activity via share-sheet, screenshot upload, or (Android) accessibility-based capture.
FR-2	On-device Pre-filtering	System shall run a lightweight on-device classifier to determine whether captured content is civic-related before any cloud call.
FR-3	OCR Text Extraction	System shall extract text from evidence using on-device OCR (ML Kit / Vision framework).
FR-4	Cloud Verification	System shall verify source authenticity, template match, and check against known fake-confirmation patterns for content that passes pre-filtering.
FR-5	Duplicate Detection	System shall reject or flag resubmissions using perceptual image hashing and text-similarity checks.
FR-6	Points Calculation	System shall award Civic Points per category using the defined point ranges, verification-tier weighting, daily/weekly caps, and streak multipliers.
FR-7	Level & Badge Assignment	System shall update the user's planetary level (Mercury → Pluto) and unlock sub-badges as point thresholds are crossed.
FR-8	Leaderboard	System shall maintain and update, in near real time, Global, Country, City, Friends, and Organization leaderboards across All-time, Monthly, and Weekly timeframes.
FR-9	Privacy Controls	User shall be able to view a transparency log of scanned content, delete it, and opt out of public leaderboards.
FR-10	Social Sharing	System shall generate shareable badge/points image cards for export to WhatsApp (Business API bot) and Instagram (Share Sheet / Stories).
FR-11	Appeals & Review	User shall be able to appeal a rejected or flagged submission; a moderator queue shall process appeals.
FR-12	Fraud Monitoring	System shall apply rate limiting, category caps, and device/account fingerprinting to detect multi-accounting and leaderboard gaming.

- **Non-Functional Requirements**

ID	Requirement	Description
NFR-1	Performance	On-device pre-filter and OCR shall complete within 3 seconds on a mid-range Android device; leaderboard updates shall propagate within 5 seconds of a verified point award.
NFR-2	Security	All evidence in transit shall be encrypted (TLS 1.2+); stored evidence shall be encrypted at rest and auto-deleted after the verification window.
NFR-3	Privacy & Compliance	System shall comply with India's DPDP Act, GDPR/CCPA where applicable, Google Play's
		Accessibility API policy, and Apple's App Store screen-recording guidelines.

NFR-4	Usability	Onboarding shall clearly explain what is and is not scanned, requiring explicit opt-in consent before any accessibility/screen-capture permission is requested.
NFR-5	Reliability	Core services (Points Engine, Verification, Leaderboard) shall target 99.5% uptime; failed verifications shall not deduct or duplicate points.
NFR-6	Scalability	Backend microservices shall scale horizontally to support city-wide or campus-wide leaderboard events without service degradation.
NFR-7	Compatibility	Client app shall support low-end/low-bandwidth Android devices as the primary target, in addition to standard iOS devices.
NFR-8	Maintainability	Backend shall be organized as independently deployable microservices to ease updates and scaling.

- **Use Cases**

ID	Use Case	Actor	Description
UC-1	Submit Civic Evidence	Registered User	User shares/uploads proof of a civic action; system extracts, verifies, and scores it.
UC-2	View Leaderboard	Registered User	User filters and views rank across scope (Global/City/Org) and timeframe (Weekly/Monthly/All-time).
UC-3	View Orbit Map / Badge Locker	Registered User	User views current planetary level, progress bar, and unlocked/locked badges.
UC-4	Share Achievement	Registered User	User exports a generated badge/points card to WhatsApp or Instagram.
UC-5	Appeal Rejected Submission	Registered User	User contests a flagged/rejected evidence submission for manual review.
UC-6	Review Flagged Submission	Moderator	Moderator inspects flagged evidence and approves, rejects, or requests more proof.

Draw system architecture, identify major modules

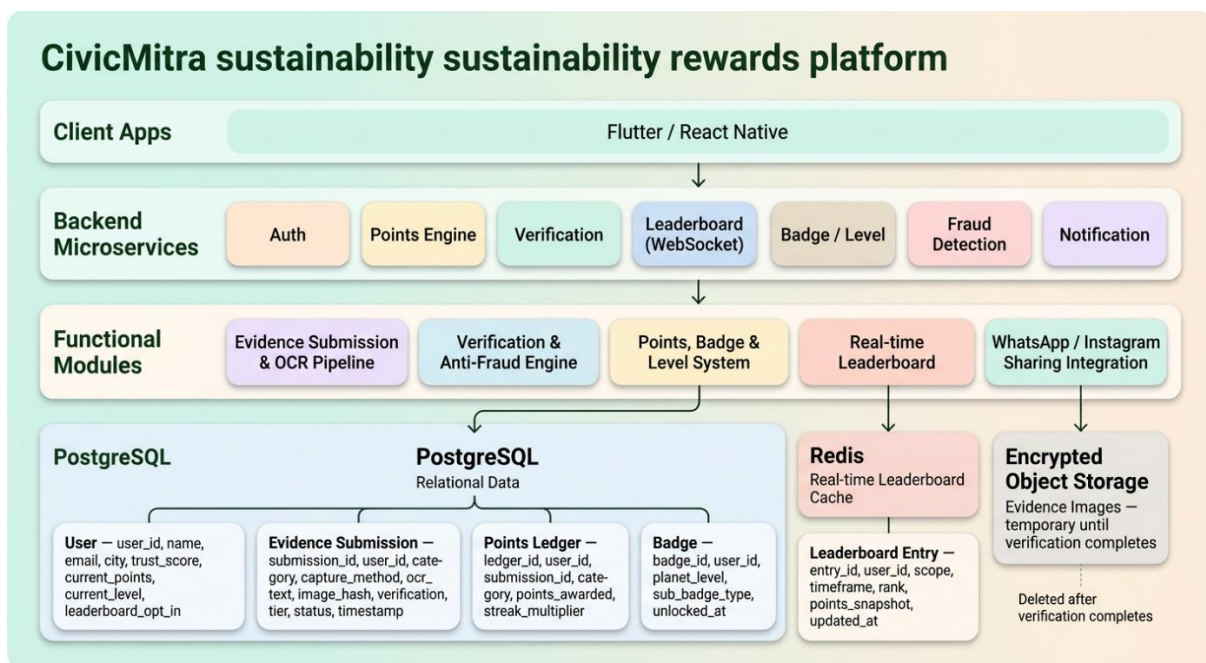
The client (Flutter/React Native) communicates with a backend composed of independent microservices: Auth, Points Engine, Verification, Leaderboard (WebSocket-based), Badge/Level, Fraud Detection, and Notification. PostgreSQL stores relational data, Redis caches real-time leaderboard state, and encrypted object storage holds evidence images only until verification completes. The major functional modules identified are:

- Evidence Submission & OCR Pipeline
- Verification & Anti-Fraud Engine

- Points, Badge & Level System
- Real-time Leaderboard
- WhatsApp / Instagram Sharing Integration

Core Data Entities

Entity	Key Fields
User	user_id, name, email, city, trust_score, current_points, current_level, leaderboard_opt_in
Evidence Submission	submission_id, user_id, category, capture_method, ocr_text, image_hash, verification_tier, status, timestamp
Points Ledger	ledger_id, user_id, submission_id, category, points_awarded, streak_multiplier, awarded_at
Badge	badge_id, user_id, planet_level, sub_badge_type, unlocked_at
Leaderboard Entry	entry_id, user_id, scope, timeframe, rank, points_snapshot, updated_at



Set up tools/environment, start basic development

- Client: Flutter / React Native, targeting Android 8.0+ and iOS 14+, optimized for low-end/low-bandwidth Android devices.
- On-device ML: TensorFlow Lite / ML Kit (Android), Core ML & Vision (iOS) for OCR and the pre-filter classifier.
- Backend: Node.js/Python microservices, containerized with Docker/Kubernetes.
- Database: PostgreSQL (relational data), Redis (leaderboard cache), S3-compatible encrypted object storage (evidence, auto-deleted post-verification).
- Messaging integrations: WhatsApp Business Platform (Cloud API) and Instagram Graph API.

Begin writing design and implementation documents

External Interface Requirements

- User Interfaces: Onboarding & consent, Home (orbit map, streak), Submit Evidence, Leaderboard, Badge Locker, Profile.
- Hardware Interfaces: Device camera and local storage for temporary evidence caching.
- Software Interfaces: ML Kit / Core ML, WhatsApp Business Platform, Instagram Graph API, Firebase/WebSocket gateway.
- Communication Interfaces: HTTPS/TLS 1.2+ for REST traffic; WSS for real-time leaderboard updates; FCM/APNs for push notifications

Implementation of core modules, integrate feedback

Development began with the points, badge, and dashboard modules, implemented and deployed under the working name CivicMitra. The current build tracks user impact (CO₂ saved, electricity saved, water saved, waste reduced), daily challenges with category-wise point values, a level/rank progress bar, and a badge collection with locked and unlocked achievement tiers. Early internal feedback on the dashboard layout and challenge presentation has been incorporated into the current build, shown in the demo screenshot under Section 9.

Perform unit and integration testing, document issues

This section defines the testing methods applied to CivicOrbit, the test cases used to validate individual modules, and the unit and integration testing performed across the system before issues are logged and tracked for resolution.

- **Testing Methods**

- **Unit Testing:** Individual functions and services (points calculation, badge threshold checks, duplicate-detection hashing, OCR parsing) are tested in isolation using JUnit/PyTest on the backend and Flutter Test/Jest on the client.
- **Integration Testing:** Validates data flow across module boundaries — e.g., Evidence Submission → OCR → Verification → Points Engine → Leaderboard — using API-level tests (Postman/Newman, Supertest) against staging services.
- **System (End-to-End) Testing:** Full user journeys are exercised on real/emulated devices, from evidence capture to leaderboard update and social sharing.
- **Regression Testing:** The full automated test suite is re-run after each bug fix or feature change to confirm no existing functionality is broken.
- **UI/Functional Testing:** Screen-level flows (submit evidence, view leaderboard, badge locker) are tested with Appium (Android) / XCUITest (iOS).
- **Black-box and White-box Testing:** Black-box testing validates functional behaviour against requirements (FR-1 to FR-12) without inspecting internal code; white-box testing is used for critical logic such as points calculation and fraud-detection rules to verify branch coverage.

- **Test Cases**

Test Case ID	Module	Test Scenario	Expected Result
TC-01	Evidence Submission	Upload a valid “I Voted” confirmation screenshot	Image accepted; OCR extracts text; content classified as civic-related
TC-02	Evidence Submission	Upload an unrelated photo (e.g., a selfie)	Pre-filter classifier rejects the image as non-civic content
TC-03	OCR Pipeline	Extract text from a municipal complaint receipt	Extracted text matches source content with acceptable accuracy
TC-04	Verification	Submit a screenshot identical to a prior submission	Duplicate detected via perceptual hashing; submission rejected
TC-05	Verification	Submit a tampered/edited confirmation image	Cloud verification flags the submission for manual moderator review
TC-06	Points Engine	Award points for a Medium-tier verified complaint	Correct point value (10–25) credited to the user's balance
TC-07	Points Engine	Submit more complaints than the daily category cap	System blocks points beyond the configured daily cap
TC-08	Badge & Level	User's total points cross the 100-point threshold	User is promoted from Mercury to Venus; badge unlocked

TC-09	Leaderboard	A verified submission awards new points	Leaderboard rank updates within 5 seconds (NFR-1)
TC-10	Leaderboard	User opts out of the public leaderboard	User is excluded from public leaderboard views but retains private progress
TC-11	Social Sharing	User shares an earned badge	A correctly rendered badge card is generated and exported to WhatsApp/Instagram
TC-12	Appeals & Review	User appeals a rejected submission	Submission enters the moderator review queue with status “Pending Appeal”
TC-13	Fraud Detection	Multiple accounts register from the same device	System flags a device-fingerprint anomaly for review
TC-14	Privacy Controls	User deletes an entry from the transparency log	Entry is permanently removed and deletion is confirmed to the user
TC-15	End-to-End Integration	Full flow: submit → verify → award points → leaderboard update	All modules update consistently within expected time, with no data loss

- **Unit and Integration Testing by Module**

Module	Unit Testing Focus	Integration Testing Focus
Evidence Submission & OCR Pipeline	Unit: validate the on-device pre-filter classifier's accept/reject decisions and OCR text-extraction accuracy on sample screenshots (TC-01, TC-02, TC-03).	Integration: confirm captured evidence is correctly handed off to the Verification service with all required metadata (category, timestamp, image hash).
Verification & Anti-Fraud Engine	Unit: test perceptual-hashing and text-similarity functions against known duplicate/near-duplicate pairs (TC-04); test template-match logic against known fake-confirmation patterns (TC-05).	Integration: confirm only submissions that pass verification are forwarded to the Points Engine, and rejected/flagged submissions correctly reach the Appeals queue (TC-12).
Points, Badge & Level System	Unit: test points calculation per category, daily/weekly cap enforcement, streak multipliers, and level-threshold logic (TC-06, TC-07, TC-08).	Integration: confirm a points award correctly triggers a badge/level check and a leaderboard update in the same transaction.
Real-time Leaderboard	Unit: test ranking and sorting logic for each scope (Global/City/Org) and timeframe (Weekly/Monthly/All-time).	Integration: confirm WebSocket/Firestore listeners propagate a new points award to connected clients within the 5-second target (TC-09, TC-10).
WhatsApp / Instagram Sharing	Unit: test badge/points card image generation for correct layout and data.	Integration: test the WhatsApp Business API bot reply flow and the Instagram Share Sheet deep
		link against sandbox/mock endpoints (TC-11).

Fraud Detection	Unit: test device/account fingerprinting rules and rate-limit thresholds.	Integration: confirm flagged accounts are correctly surfaced in the moderator review queue and do not affect other users' leaderboard standings (TC-13).
Privacy & Transparency Controls	Unit: test transparency-log entry creation and deletion logic.	Integration: confirm a deleted log entry is also removed from any cached/derived views (TC-14).

- **Reliability and Performance Targets**

The following non-functional targets guide pass/fail criteria during testing; issues found are logged and tracked for resolution before the next iteration.

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NFR-6	Scalability	Backend microservices shall scale horizontally to support city-wide or campus-wide leaderboard events without service degradation.

Rehearse demo, refine UI, prepare slides or posters

The screenshot below shows the current CivicMitra dashboard used for demo rehearsal: it surfaces the user's impact metrics, daily challenges, level/rank progress, and badge collection — the core surfaces used while refining the UI ahead of slides and the project demonstration.

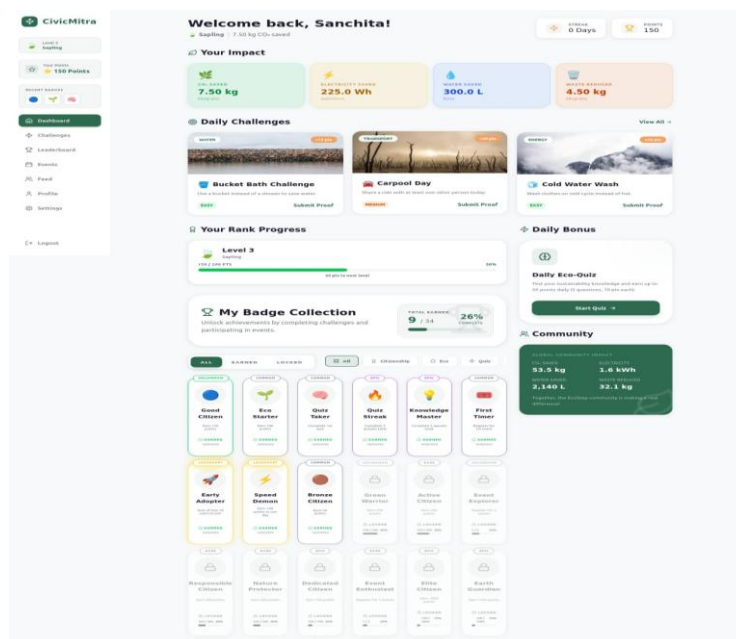


Fig. 1: CivicMitra dashboard — impact metrics, daily challenges, and rank progress.

Final report writing, include ethical impact statement

Ethical Impact Statement

CivicOrbit is designed with a privacy-first and consent-driven approach. Screen content is processed on-device wherever possible, and raw evidence is retained only for a short verification window before deletion. Users are given a visible transparency log of everything that has been scanned, with the ability to review and delete it at any time, and can opt out of public leaderboards while still tracking personal progress privately. The verification-tier and anti-fraud design (rate limiting, duplicate detection, moderator review) aims to keep the points and leaderboard system fair rather than exploitable. The system is built to comply with India's Digital Personal Data Protection (DPDP) Act, and where applicable GDPR/CCPA, as well as Google Play's Accessibility API policy and Apple's App Store guidelines on screen recording, so that civic-engagement tracking does not come at the cost of user privacy or platform trust.

Final demonstration, submission of report and logbook

The final phase of the project involves a live demonstration of the CivicOrbit/CivicMitra application, followed by submission of the complete project report, this SRS, and the maintained project logbook to the department for evaluation.

References

Aligned with SDG 11 — Sustainable Cities and Communities

- [1] M. Romano, P. Díaz, and I. Aedo, “Gamification-less: May gamification really foster civic participation? A controlled field experiment,” *Journal of Ambient Intelligence and Humanized Computing*, 2021. <https://doi.org/10.1007/s12652-021-03322-6>
- [2] A. Vanolo, “Playable Urban Citizenship: Social Justice and the Gamification of Civic Life,” 2019. <https://doi.org/10.1108/978-1-78769-139-120191004>

Aligned with SDG 12 — Responsible Consumption and Production

- [3] G. Guillén, J. Hamari, and J. Quist, “Gamification of Sustainable Consumption: A Systematic Literature Review,” *Proc. 54th Hawaii International Conference on System Sciences (HICSS)*, pp. 1345–1354, 2021. <https://doi.org/10.24251/HICSS.2021.163>
- [4] M. Nkwo, B. Suruliraj, and R. Orji, “Persuasive Apps for Sustainable Waste Management: A Comparative Systematic Evaluation of Behavior Change Strategies and State-of-the-Art,” *Frontiers in Artificial Intelligence*, 2021. <https://doi.org/10.3389/frai.2021.748454>

Aligned with SDG 16 — Peace, Justice and Strong Institutions

- [5] S. Lubis, E. P. Purnomo, J. A. Lado, and C.-F. Hung, “Electronic governance in advancing sustainable development goals through systematic literature review,” *Discover Global Society*, vol. 2, no. 1, pp. 1–22, 2024. <https://doi.org/10.1007/s44282-024-00102-3>
- [6] A. Sadat, “Digital governance and civic inclusion to enhance public participation in political decision-making processes,” *Frontiers in Political Science*, vol. 7, 2025. <https://doi.org/10.3389/fpos.2025.1671373>

Aligned with SDG 17 — Partnerships for the Goals

- [7] X. Quan and M. C. Solheim, “Public-private partnerships in smart cities: A critical survey and research agenda,” *City, Culture and Society*, vol. 32, 100491, 2023. <https://doi.org/10.1016/j.ccs.2022.100491>
- [8] G. Siokas, V. Kelaidi, and A. Tsakanikas, “The smart city as a hub for nourishing public-private partnerships,” *Sustainable Cities and Society*, vol. 76, 103466, 2022. <https://doi.org/10.1016/j.scs.2021.103466>